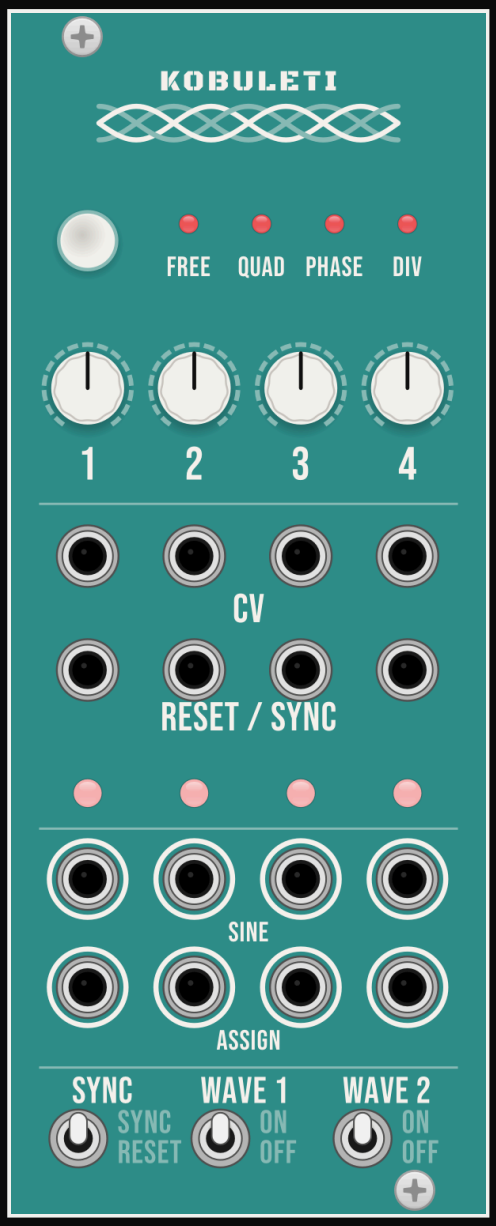
BETOBULLET

10 HP · EURORACK FORMAT
FOR VCV RACK 2



KOBULETI

QUADRATURE / PHASE / DIVIDE LFO

USER MANUAL · v2.0.0 · October 2026

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Kobuleti is a quadruple LFO with four ways of relating its channels: free running, quadrature, phase-shifted and divided. Every channel has a sine output and a second output whose waveform you choose.

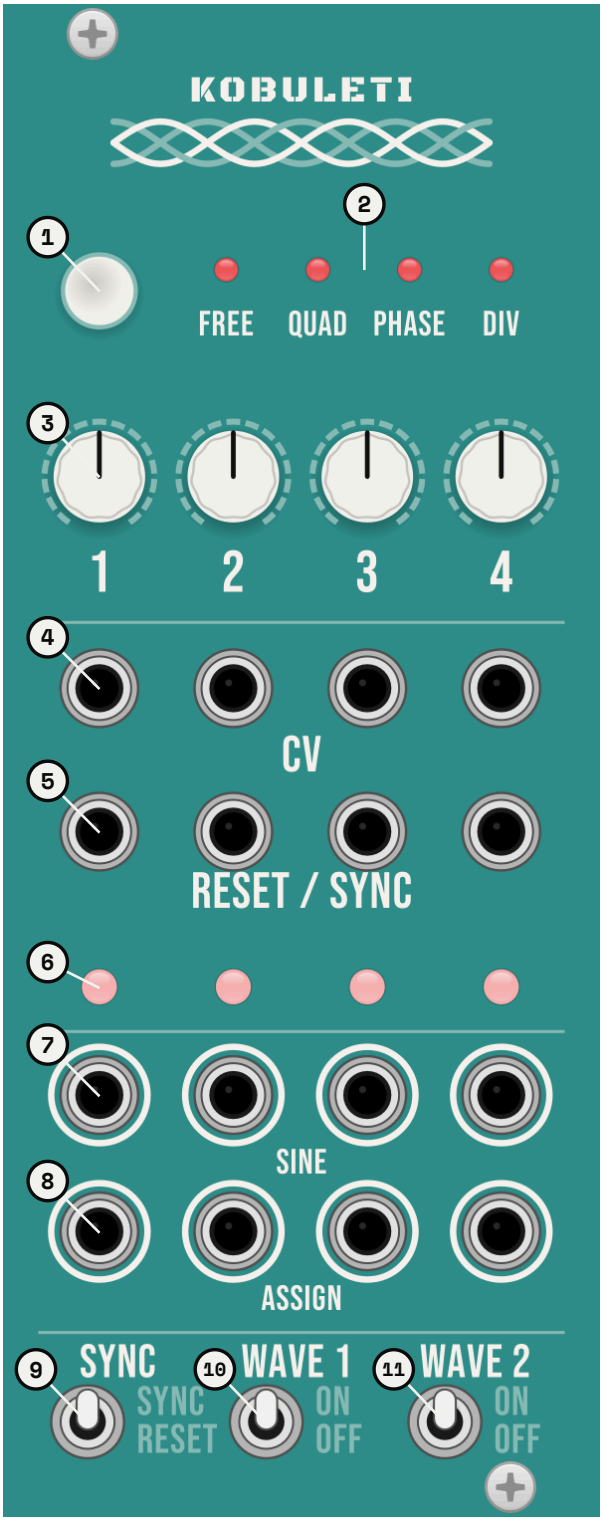
1. INTRODUCTION

Kobuleti runs the open source firmware of the Xaoc Devices Batumi (MIT licence, Matthias Puech) inside VCV Rack. The LFO engine (lfo, processor and resources) is unchanged and runs at the hardware's own rate of 16384 Hz, so frequencies, waveshapes, sync, reset and divisions are exactly those of the original. Only the input/output layer was rewritten for the Rack.

INSTALLATION

Copy `Kobuleti-2.0.0-win-x64.vcvplugin` to `Documents/Rack2/plugins` and restart VCV Rack. The module is listed under the brand **betobullet**.

2. PANEL OVERVIEW



1. **SELECT**
   Short press: next mode. Hold 0.5 s: fine tune (zoom).
2. **Mode LEDs**
   FREE, QUAD, PHASE, DIV.
3. **Knobs 1–4**
   Frequency, or the parameter of the current mode.
4. **CV 1–4**
   Frequency CV, 1 V/octave.
5. **RESET / SYNC 1–4**
   Reset or sync triggers, depending on switch (9).
6. **Channel LEDs**
   White on the positive half of the wave, grey on the negative half.
7. **SINE 1–4**
   Sine outputs, ±5 V.
8. **ASSIGN 1–4**
   Outputs with the waveform chosen by switches (10) and (11).
9. **SYNC / RESET**
   What the RESET inputs do: restart the phase or synchronise the period.
10. **WAVE 1**
    Waveform of the ASSIGN outputs, with WAVE 2.
11. **WAVE 2**
    Together: trapezoid, ramp, saw or triangle.

3. MODES

A short press on SELECT **(1)** steps through the four modes; the mode LEDs **(2)** show the current one.

MODE	KNOB 1	KNOBS 2–4
FREE	frequency of channel 1	frequency of channels 2–4: four independent LFOs
QUAD	common frequency	level of each channel; phases fixed at 0°, 90°, 180°, 270°
PHASE	common frequency	phase shift of each channel relative to channel 1
DIV	common frequency	integer division of the frequency for each channel

INPUTS IN QUAD, PHASE AND DIV

When the channels share one frequency, the RESET inputs of channels 2–4 take new jobs:

- **RESET 2** freezes the LFO.
- **RESET 3** reverses the direction.
- **RESET 4** steps to the next waveform.

FINE TUNE (ZOOM)

Hold SELECT for half a second to enter fine tune: the knobs now adjust in small steps around their position and the mode LED blinks. A short press leaves fine tune. If a knob was moved while zoomed, its LED keeps blinking until the knob returns to its old position, so nothing jumps.

4. OUTPUTS AND SWITCHES

Each channel has a sine output **(7)** and an ASSIGN output **(8)**. The two WAVE switches **(10, 11)** choose the waveform of all ASSIGN outputs:

WAVE 1	WAVE 2	ASSIGN OUTPUT
off	off	trapezoid
on	off	ramp
off	on	saw
on	on	triangle

The SYNC/RESET switch **(9)** sets what the RESET inputs **(5)** do: **RESET** restarts the cycle at every trigger; **SYNC** measures the time between triggers and locks the LFO period to it.

PATCH IDEAS

- **Quadrature panning**: QUAD mode, SINE 1 and SINE 2 to the pan inputs of two mixer channels.
- **Polyrhythmic modulation**: DIV mode with a clock into RESET 1 in SYNC; the divisions stay locked.
- **Moving phaser stages**: PHASE mode, four outputs into four filter cutoffs.

5. SPECIFICATIONS

ITEM	VALUE
Width	10 HP
Channels	4
Modes	FREE, QUAD, PHASE, DIV
Engine rate	16384 Hz (as hardware)
Frequency CV	1 V/octave
Outputs	4 × sine, 4 × assignable, ±5 V
Reset / sync threshold	about 3 V

CREDITS

LFO engine: Batumi firmware by Matthias Puech (MIT). Port, panel and VCV layer: betobullet. Batumi is a product of Xaoc Devices. Fonts: Black Ops One, Space Mono, Space Grotesk (SIL OFL).